

Attention Is All You Need

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Motivation & Background

- Recurrent and convolutional models dominate sequence transduction tasks
- Attention mechanisms improve modeling of long-range dependencies
- Sequential computation in RNNs limits parallelization and efficiency
- Transformer replaces recurrence and convolutions with attention
- Enables faster training and improved translation quality

Transformer Model Architecture

- Encoder-decoder structure with stacked self-attention and feed-forward layers
- Dispenses with recurrence and convolutions entirely
- Enables highly parallel computation
- Residual connections and layer normalization in each sub-layer

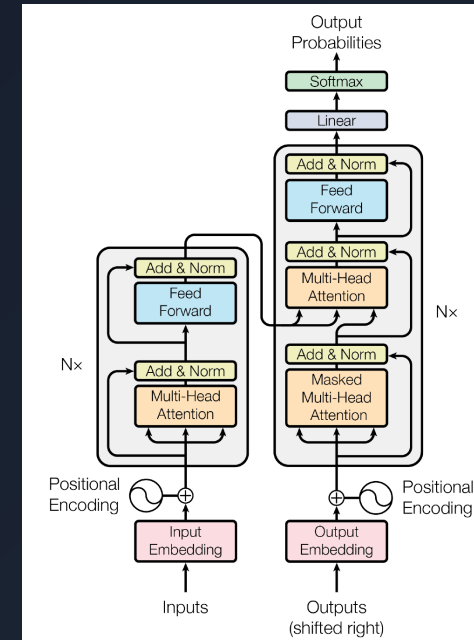


Figure 1: The Transformer model architecture.

Attention Mechanisms in Transformer

- Scaled Dot-Product Attention: computes attention weights via scaled dot products
- Multi-Head Attention: parallel attention layers capture diverse relationships
- Allows model to attend to information from different subspaces

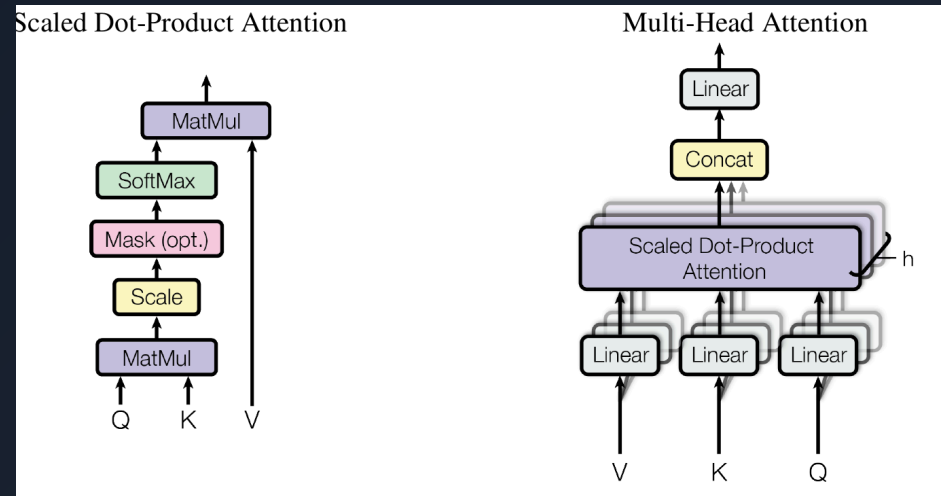


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

Advantages of Self-Attention

- Shorter paths for long-range dependencies than RNNs or CNNs
- Enables full parallelization during training
- Computationally efficient for typical sequence lengths
- Facilitates interpretability via attention visualization

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations for different layer types. n is the sequence length, d is the representation dimension, k is the kernel size of convolutions and r the size of the neighborhood in restricted self-attention.

Key Components of the Transformer

- Encoder: stack of 6 layers, each with multi-head self-attention and feed-forward network
- Decoder: stack of 6 layers, includes encoder-decoder attention
- Positional encodings added to input embeddings to capture order
- Shared embedding and pre-softmax weights

Training Regime

- Trained on WMT 2014 English-German (4.5M pairs) and English-French (36M pairs)
- Byte-pair encoding for vocabulary
- Adam optimizer with custom learning rate schedule
- Dropout and label smoothing for regularization
- Base model: 12 hours, Big model: 3.5 days on 8 GPUs

Translation Results & Efficiency

- Transformer outperforms previous models on BLEU scores
- Achieves higher quality at a fraction of training cost
- Base and Big models both surpass prior single and ensemble models

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model Variations & Ablations

- Performance evaluated under various architectural changes
- Increasing model size and dropout improves BLEU
- Learned vs. sinusoidal positional encodings yield similar results

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ts}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$		
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65		
(A)					1	512	512				5.29	24.9		
					4	128	128				5.00	25.5		
					16	32	32				4.91	25.8		
					32	16	16				5.01	25.4		
(B)					16						5.16	25.1	58	
					32						5.01	25.4	60	
(C)	2											6.11	23.7	36
	4											5.19	25.3	50
	8											4.88	25.5	80
			256			32	32				5.75	24.5	28	
			1024			128	128				4.66	26.0	168	
			1024								5.12	25.4	53	
			4096								4.75	26.2	90	
(D)							0.0			5.77	24.6			
							0.2			4.95	25.5			
									0.0	4.67	25.3			
									0.2	5.47	25.7			
(E)	positional embedding instead of sinusoids									4.92	25.7			
big	6	1024	4096	16					0.3	300K	4.33	26.4	213	

Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base model, All metrics are on the English-to-German translation development set, newstest2013. Listed perplexities are per-wordpiece, according to our byte-pair encoding, and should not be compared to per-word perplexities.

Generalization: Constituency Parsing

- Transformer achieves strong results on English constituency parsing
- Outperforms prior models in both small and large data regimes
- Minimal task-specific tuning required

Parser	Training	WSJ 23 F1
Vinyals & Kaiser et al. (2014) [37]	WSJ only, discriminative	88.3
Petrov et al. (2006) [29]	WSJ only, discriminative	90.4
Zhu et al. (2013) [40]	WSJ only, discriminative	90.4
Dyer et al. (2016) [8]	WSJ only, discriminative	91.7
Transformer (4 layers)	WSJ only, discriminative	91.3
Zhu et al. (2013) [40]	semi-supervised	91.3
Huang & Harper (2009) [14]	semi-supervised	91.3
McClosky et al. (2006) [26]	semi-supervised	92.1
Vinyals & Kaiser et al. (2014) [37]	semi-supervised	92.1
Transformer (4 layers)	semi-supervised	92.7
Luong et al. (2015) [23]	multi-task	93.0
Dyer et al. (2016) [8]	generative	93.3

Table 4: The Transformer generalizes well to English constituency parsing (Results are on Section 23 of WSJ)

Attention Visualization: Long-Range Dependencies

- Attention heads capture long-distance dependencies
- Example: heads attend to distant verb-object relationships
- Visualizations reveal interpretable model behavior

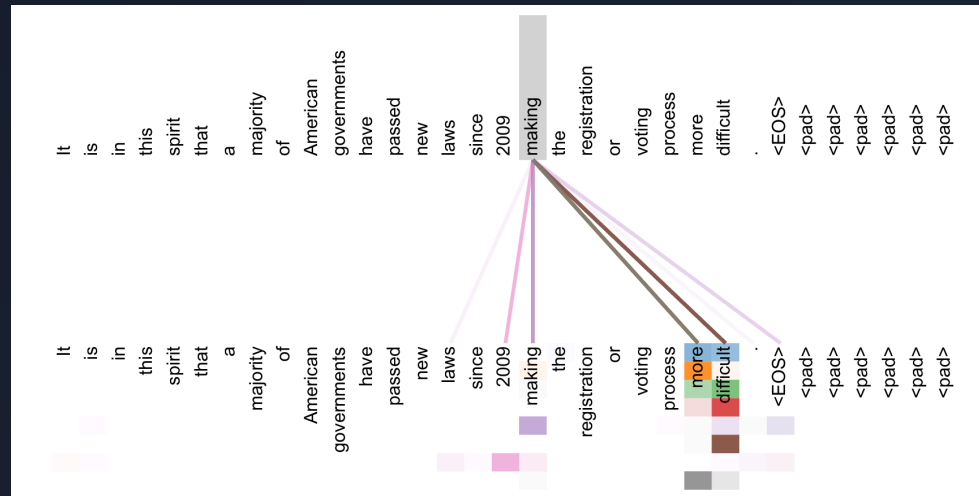


Figure 3: An example of the attention mechanism following long-distance dependencies in the encoder self-attention in layer 5 of 6. Many of the attention heads attend to a distant dependency of the verb 'making', completing the phrase 'making...more difficult'. Attentions here shown only for the word 'making'. Different colors represent different heads. Best viewed in color.

Attention Visualization: Anaphora Resolution

- Specific heads focus on pronoun resolution (e.g., 'its')
- Sharp attention distributions indicate precise reference tracking
- Demonstrates specialization among attention heads



Figure 4: Two attention heads, also in layer 5 of 6, apparently involved in anaphora resolution. Top: Full attentions for head 5. Bottom: Isolated attentions from just the word 'its' for attention heads 5 and 6. Note that the attentions are very sharp for this word.

Attention Heads Specialization

- Different heads learn distinct syntactic and semantic tasks
- Visualization shows heads capturing sentence structure
- Supports model interpretability and analysis

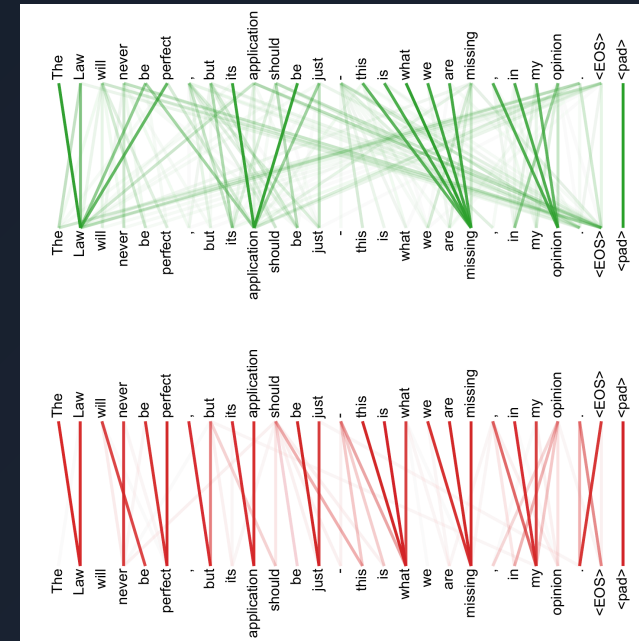


Figure 5: Many of the attention heads exhibit behaviour that seems related to the structure of the sentence. We give two such examples above, from two different heads from the encoder self-attention at layer 5 of 6. The heads clearly learned to perform different tasks.

Conclusion & Future Work

- Transformer: first sequence model based solely on attention
- Achieves state-of-the-art results in translation and parsing
- Enables faster, more parallel training
- Future: extend to other modalities and efficient local attention